



Assignment Title:

Coursework

Coursework Type:

Individual

Module Name:

<ST4005CEM Database System>

Intake: September

Submitted By

CU ID: 12332650
College ID: 210310
Name: Aditya Chaulagain

Submitted To

Name: Hira Datta Dhaka

Table of Contents

Introduction.....	4
Normalization.....	5
Entity Relationship diagram.....	8
SQL Queries	9
Data Visualization.....	14
Conclusion	20
References.....	21

Table of Figure

Figure 1 First Normal Form	5
Figure 2 Second Normal Form	6
Figure 3 Third Normal Form.....	7
Figure 4 ER Diagram	8
Figure 5 Creating Database	9
Figure 6 User Table created	10
Figure 7 Order Table Created.....	11
Figure 8 Bill Table Created	12
Figure 9 Product Table Created	13
Figure 10 Supplier_Product	14
Figure 11 Table join.....	14
Figure 12 Libraries.....	15
Figure 13 Manipulating Data	15
Figure 14 Counting, null value	16
Figure 15 Delete Null value	16
Figure 16 Bar Graph	17
Figure 17 Histogram.....	18
Figure 18 Pie chart	19

Introduction

A database is a planned gathering of data. They enable the manipulation and storage of data electronically. Data administration is made simple by databases.

Let's use a database as an example. A database is used to hold information on people, their phone numbers, and other contact information in an online telephone directory. A database is used by your energy service provider to handle invoicing, client-related issues, defect data, etc.

Let's think about Facebook as well. Data on members, their friends, member activities, messages, adverts, and much more must be stored, modified, and presented. We have endless examples of how to use databases at our disposal. What kind of organizations are needed and benefited by databases? The quick answer is any company or organization that must manage a big volume of clients or goods. By "big," we imply much more than what the human brain is capable of holding. At this juncture, a skeptic may still contend that there are still plenty of mom-and-pop businesses whose proprietors manage their inventory and profit/loss using a trusted ledger and calculator. That's accurate, however even for extremely tiny firms, using an electronic database may be beneficial. For instance, if the store were to raise the price of ballpoint pens, a ledger cannot perform a simulation to extrapolate earnings.

Type of DBMS (database management system)

- Non-relational
- Relational

Normalization

First Normal Form

Figure 1 First Normal Form

[illegible]

Second Normal Form

Figure 2 Second Normal Form

		2nf	
user	Customer	Supplier	
username	CustId *	supplierid	
userid	Custname	companyname	
	custAddress	Contname	
	custContact	ContAddress	
		contContact	
Order	Bill	Product	
Date	BillNO*	productid	
Number	CustId	prodname	
quantity	OrderNumber	prodSize	
Order total amount	Bill Total amount	unitprice	

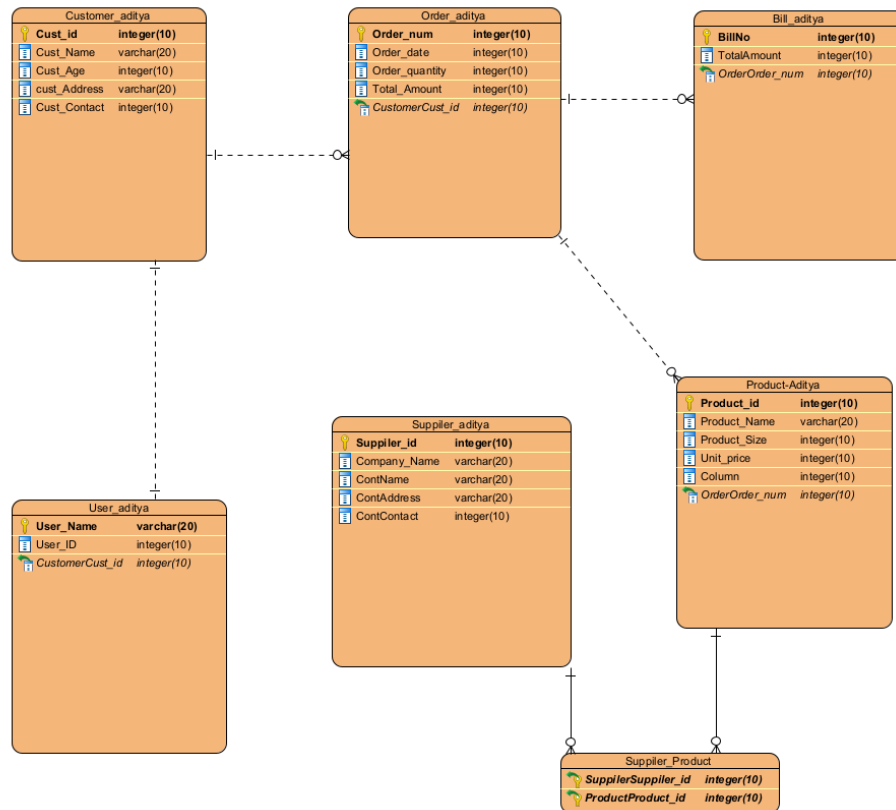
Third Normal Form

Figure 3 Third Normal Form

3NF		
user username userid	Customer CustId * Custname custAddress custContact CustAge	Supplier supplierid companyname Contname ContAddress contContact
Order Date Number quantity Order total amount	Bill BillNO* CustId OrderNumber Bill Total amount	Product productid prodname prodSize unitprice
	Supplier Product <u>Supplier_Productid*</u>	

Entity Relationship diagram

Figure 4 ER Diagram



SQL Queries

Creating Database

Figure 5 Creating Database

```
1 • create database Multination;
2 • use Multination;
3 • create table customer(
4   Cust_id int(10) not null,
5   Cust_Name varchar(20) not null,
6   Cust_Age int(10) not null,
7   Cust_Address varchar(20) not null,
8   Cust_Contact varchar(10)not null,
9   constraint Customer_pk primary key(Cust_id)
10 );
11 • insert into customer values(12,"Aditya",21,"sitapiala",9869021446);
12 • insert into customer values(13,"raju",29,"kalanki",9869021455);
13 • insert into customer values(14,"pandu",11,"maitidevi",9869021356);
14 • select *from customer;
```

Result Grid | | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	Cust_id	Cust_Name	Cust_Age	Cust_Address	Cust_Contact
▶	12	Aditya	21	sitapiala	9869021446
	13	raju	29	kalanki	9869021455
	14	pandu	11	maitidevi	9869021356
*	NULL	NULL	NULL	NULL	NULL

User Table created with value inserted

Figure 6 User Table created

```
16 • create table User(  
17     user_id int(10) not null,  
18     username varchar(20) not null,  
19     password varchar(50) not null,  
20     Cust_id int(10) not null,  
21     constraint usser_pk primary key(user_id),  
22     constraint user_fk foreign key(Cust_id) references customer(Cust_id)  
23 );  
24  
25 • insert into user values(1,"lil_pasa",md5("y23"),12);  
26 • insert into user values(2,"raj",md5("qw4"),13);  
27 • insert into user values(3,"panda123",md5("qr87"),14);  
28  
29 • select * from user;  
30
```

Result Grid				
Filter Rows:				
Edit:				
Export/Import:				
Wrap Cell Content:				
	user_id	username	password	Cust_id
▶	1	lil_pasa	1f3ec243c28dd2df13578c2d27c50541	12
	2	raj	123c0e6e77e8bcc9ea1943f9e064782	13
	3	panda123	ee53c59dffa97cfa52693d81a6faa2e2	14
*	NULL	NULL	NULL	NULL

Order table created with value inserted

Figure 7 Order Table Created

```
31 • create table orders(  
32     order_num int(10) not null,  
33     order_date date not null,  
34     order_quantity int(10) not null,  
35     Total_Amount int(10) not null,  
36     Cust_id int(10) not null,  
37     constraint orders_pk primary key(order_num),  
38     constraint orders_fk foreign key(Cust_id) references customer(Cust_id)  
39 );  
40  
41 • insert into orders values(1, '2021-12-2', 5, 100, 12);  
42 • insert into orders values(2, '2021-12-3', 4, 130, 13);  
43 • insert into orders values(3, '2021-12-5', 3, 120, 14);  
44  
45 • select * from orders;
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	order_num	order_date	order_quantity	Total_Amount	Cust_id
▶	1	2021-12-02	5	100	12
	2	2021-12-03	4	130	13
	3	2021-12-05	3	120	14
*	NULL	NULL	NULL	NULL	NULL

Bill table created

Figure 8 Bill Table Created

```
60 • create table bill(  
61     BillNo int(10) not null,  
62     TotalAmount int(10) not null,  
63     order_num int(10) not null,  
64     constraint Bill_pk primary key(BillNo),  
65     constraint Bill_fk foreign key(Order_num) references Orders(Order_num)  
66 );  
67 • insert into bill values(1,2,1);  
68 • insert into bill values(2,3,2);  
69 • select * from bill;  
70  
71 • create table Product(  
72     Product_id int(10) not null,
```

Result Grid			
Filter Rows:			
Edit: Export/Import: Wrap Cell Content:			
	BillNo	TotalAmount	order_num
▶	1	2	1
	2	3	2
*	NULL	NULL	NULL

Product table created

Figure 9 Product Table Created

```
71 • create table Product(  
72     Product_id int(10) not null,  
73     Product_Name varchar(20) not null,  
74     Product_Size int(10) not null,  
75     Unit_price int(10) not null,  
76     Order_num int(10) not null,  
77     constraint Product_pk primary key(Product_id)  
78 );  
79 • insert into Product values(1,"apple",20,400,10);  
80 • insert into Product values(2,"ball",30,500,20);  
81 • insert into Product values(3,"cat",40,600,30);  
82  
83 • select * from Product;
```

Result Grid		Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
	Product_id	Product_Name	Product_Size	Unit_price	Order_num
▶	1	apple	20	400	10
	2	ball	30	500	20
	3	cat	40	600	30
*	NULL	NULL	NULL	NULL	NULL

Supplier product created

Figure 10 Supplier_Product

```
85 • create table Supplier_product(  
86     Supplier_id int(10) not null,  
87     Product_id int(10) not null,  
88     constraint table_Pk primary key(Supplier_id,Product_id),  
89     constraint Fk1 foreign key(Supplier_id) references Supplier(Supplier_id),  
90     constraint Fk2 foreign key(Product_id) references Product(Product_id)  
91 );  
92 • insert into Supplier_product values(1,1);  
93 • insert into Supplier_product values(2,2);  
94 • select * from Supplier_product;  
95
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Supplier_id	Product_id
1	1	1
2	2	2
*	NULL	NULL

Joining Table

Figure 11 Table join

```
95 • select* from orders  
96 left join bill  
97 on orders.Order_num=bill.Order_num;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	order_num	order_date	order_quantity	Total_Amount	Cust_id	BillNo	TotalAmount	order_num
▶	1	2021-12-02	5	100	12	1	2	1
	2	2021-12-03	4	130	13	2	3	2
	3	2021-12-05	3	120	14	NULL	NULL	NULL

Data Visualization

Importing libraries

Figure 12 Libraries

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Manipulating Data

Figure 13 Manipulating Data

```
In [20]: path = "C:\\Users\\asus\\Desktop\\Database\\tips.csv"
tip_1 = pd.read_csv(path)
tip_1
```

Out[20]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

Counting and checking null values

Figure 14 Counting, null value

```
In [21]: tip_1.count()
```

```
Out[21]: total_bill    244  
         tip          244  
         sex          244  
         smoker       244  
         day          244  
         time          244  
         size         244  
         dtype: int64
```

```
In [22]: tip_1.isnull().sum()
```

```
Out[22]: total_bill    0  
         tip          0  
         sex          0  
         smoker       0  
         day          0  
         time          0  
         size         0  
         dtype: int64
```

Deleting null values

Figure 15 Delete Null value

```
In [23]: tip_1.dropna(inplace=True)  
         tip_1.isnull().sum()
```

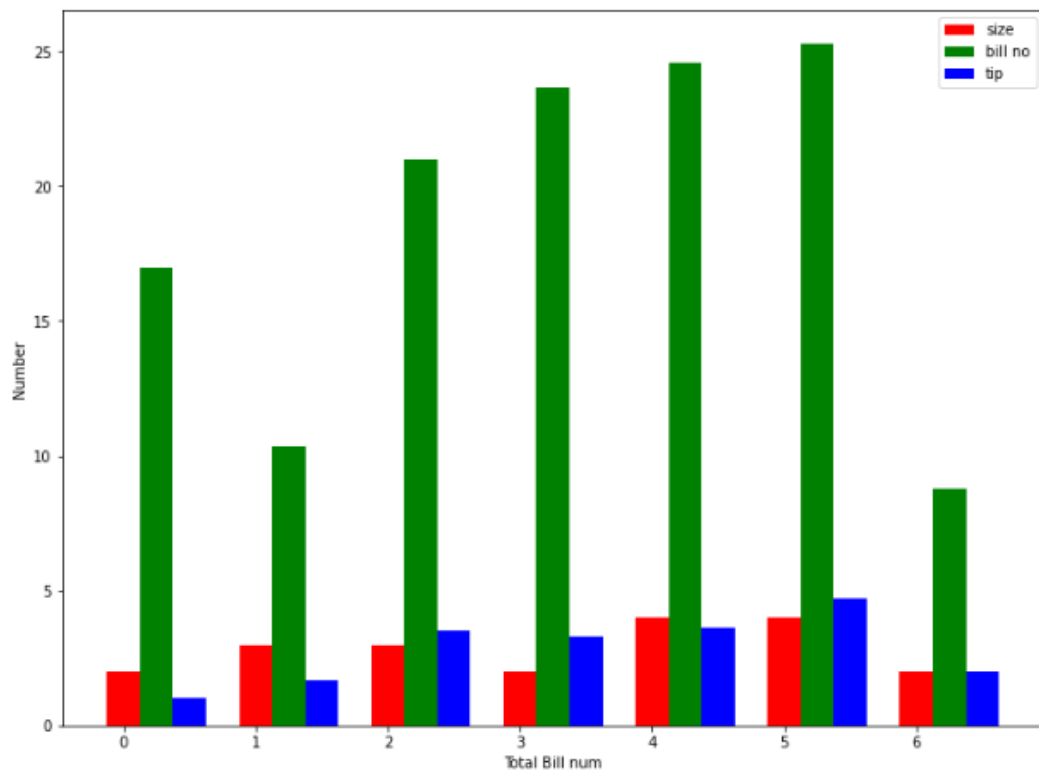
```
Out[23]: total_bill    0  
         tip          0  
         sex          0  
         smoker       0  
         day          0  
         time          0  
         size         0  
         dtype: int64
```


Bar graph

Figure 16 Bar Graph

```
In [30]: number_of_lines=7;
total_bill = tip_1["total_bill"][0:number_of_lines]
tips =tip_1["tip"][0:number_of_lines]
size = tip_1["size"][0:number_of_lines]

sizes=(12,9)
plt.figure(figsize=sizes)
number=np.arange(number_of_lines)
width=0.25
plt.bar(number,size, width, label="size",color = 'r')
plt.bar(number+width,total_bill , width, label="bill no",color = 'g')
plt.bar(number+width*2,tips, width, label="tip",color = 'b')
plt.xlabel("Total Bill num")
plt.ylabel("Number")
plt.legend()
plt.show()
```

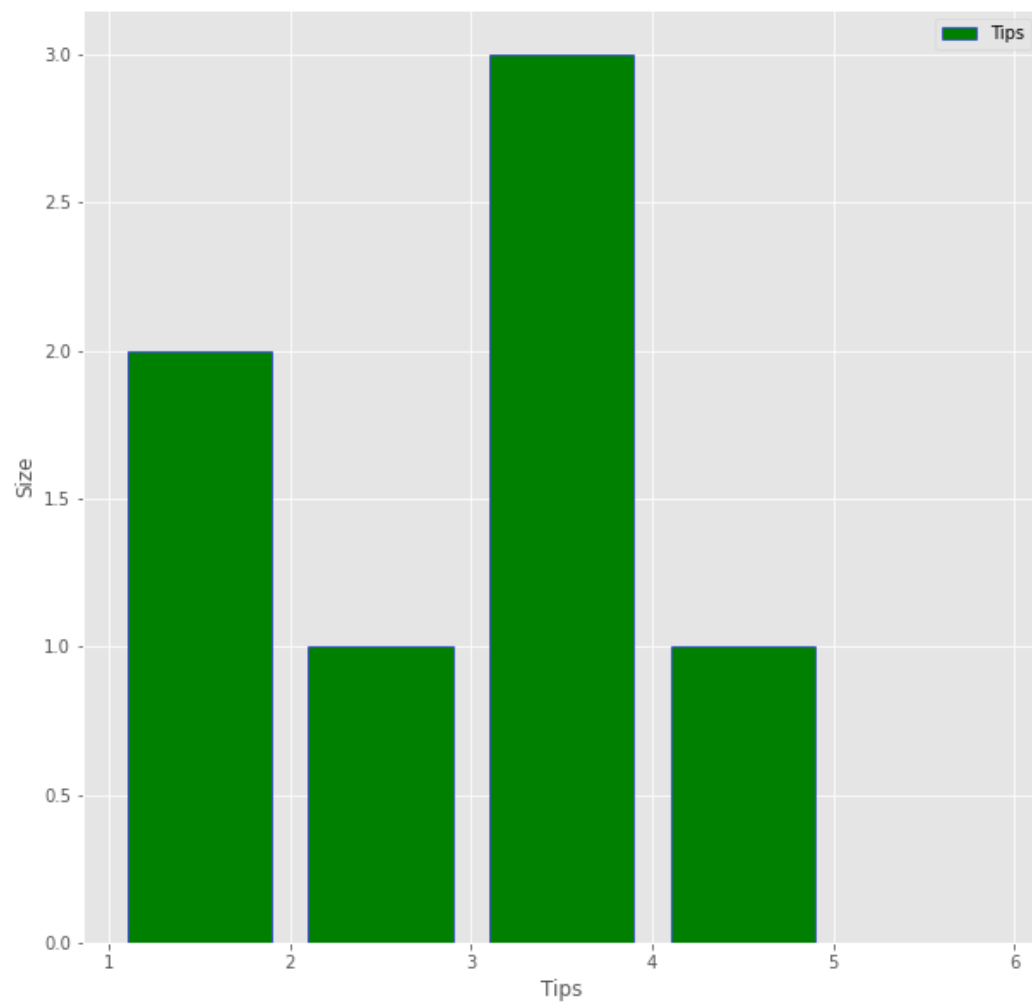


Histogram

Figure 17 Histogram

Histogram ¶

```
In [57]: from matplotlib import style
plt.figure(figsize=size)
sizes = [1,2,3,4,5,6]
plt.hist(tips,sizes,rwidth=0.8,color="g",edgecolor="b",label="Tips")
plt.xlabel("Tips")
plt.ylabel("Size")
style.use("ggplot")
plt.legend()
plt.show()
```



Pie chart

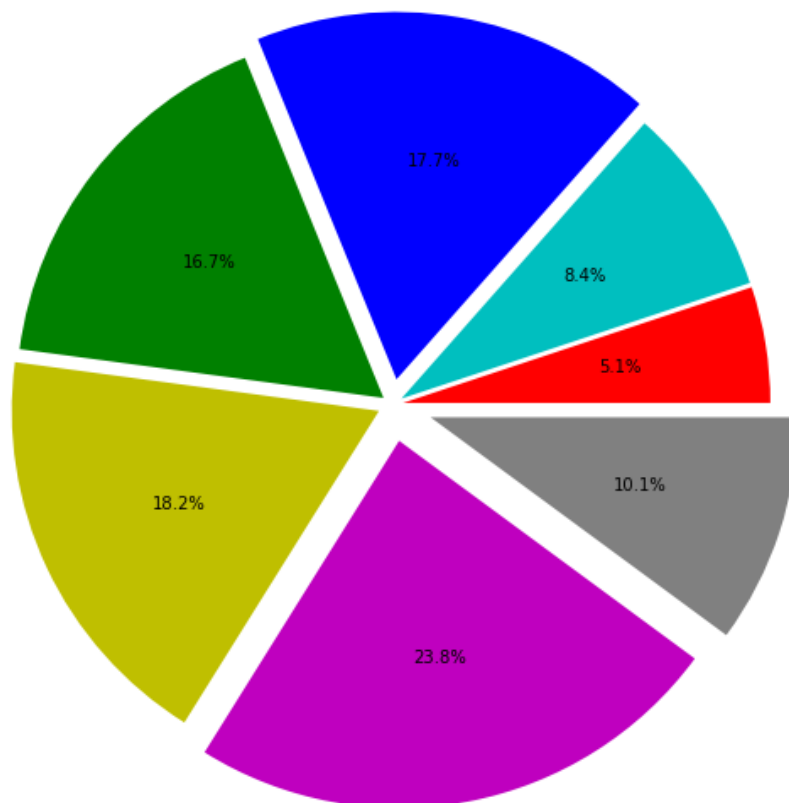
Figure 18Pie chart

Pie Chart

In []:

```
In [55]: size = (10,10)
plt.figure(figsize=size)
plt.pie(tips[0:8],
        colors=['r','c','b','g','y','m','0.5'],
        explode=[0.03,0.03,0.07,0.03,0.04,0.1,0.11,],
        autopct="%0.1f%%")
plt.title("Tips Ratio")
plt.show()
```

Tips Ratio



Conclusion

Lastly, the given assignment was all about solving the given problem with the help of database software. Er diagram was done by using visual paradigm. SQL problem was solved by using My SQL workbench, and last data visualization was done on anaconda with the help of NumPy and pandas.

References

What is a Database? Definition, Meaning, Types with Example. (2022). Retrieved 24 July 2022, from <https://www.guru99.com/introduction-to-database-sql.html>

Introduction to Databases and SQL (With Examples). (2022). Retrieved 24 July 2022, from <https://www.programiz.com/sql/database-introduction>